Health Economics

Md Sariful Islam, PhD

Associate Professor, Economics Discipline

Khulna University

2026

Contents

1 Health Policy **1**
- 1.1 Scarcity, financing, provision, and allocation . . . 1
- 1.2 The patient price, demand, and capacity . . . 2
- 1.3 The efficient amount of care . . . 3
- 1.4 Social insurance and risk aversion . . . 4
- 1.5 Population aging and demographic indicators . . . 6
- 1.6 Health-system budgets and financial sustainability . . . 8
- 1.7 Aging and Bismarck insurance contributions . . . 10

2 Economic Epidemiology and the SIR Model **12**
- 2.1 Why economists study infection dynamics . . . 12
- 2.2 SIR compartments and the closed-population assumption . . . 12
- 2.3 Derivation of infection and recovery flows . . . 13
- 2.4 Constructing the three differential equations . . . 14
- 2.5 Reproduction numbers, early growth, and epidemic peaks . . . 15
- 2.6 Reading epidemic curves and calculating an example . . . 17
- 2.7 Prevention and infectious-disease externalities . . . 19
- 2.8 Hospital congestion and financing pressures . . . 19

3 Pharmaceutical Innovation **21**
- 3.1 Pharmaceutical research as an economic activity . . . 21
- 3.2 The life cycle of a pharmaceutical drug . . . 21
- 3.3 Why pharmaceutical development is costly and uncertain . . . 23
- 3.4 Patents, prices, and access . . . 25
- 3.5 Induced innovation and the market-size incentive . . . 28
- 3.6 Drug regulation as an information problem . . . 29
- 3.7 Medical technology, effectiveness, and prices . . . 30
- 3.8 Technology and aggregate medical expenditure . . . 31
- 3.9 Cost effectiveness, budget impact, and policy design . . . 32

CHAPTER 1

Health Policy

1.1 Scarcity, financing, provision, and allocation

Question 1.1. What fundamental economic problem that every health-care system must try to solve?

Health-care needs may be very large, but trained staff, hospital beds, time, medicines, diagnostic equipment and public or private funds are limited. Every system therefore faces scarcity and opportunity cost. To increase one service, society often must use resources that could have produced another service or nonmedical consumption. A sound analysis separates three questions: *financing* (who pays), *provision* (who treats), and *allocation* (who receives care and when). Insurance or tax funding may change who pays the bill; neither eliminates the real resources needed for treatment.

Question 1.2. Distinguish between the main financing arrangements of Beveridge-type and Bismarck-type health-care systems.

Beveridge-type: Services are financed mainly from general tax revenue. Government has a central financing role and often directly runs a substantial part of the delivery system. Eligible patients frequently pay little or nothing when receiving covered care.

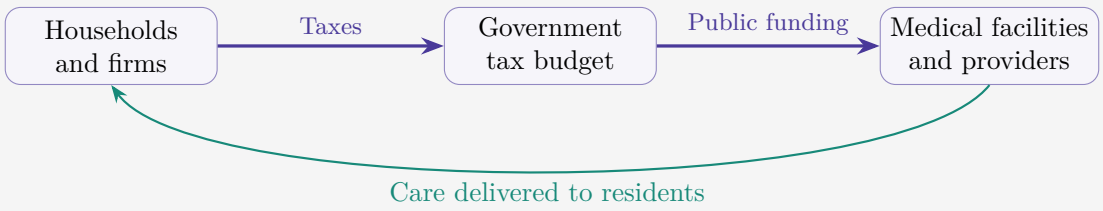


Figure 1.1: Beveridge financing: taxes support providers and care reaches residents.

Bismarck-type: Services are financed mainly through compulsory social health-insurance contributions, conventionally related to earnings and often paid by workers and employers. Insurance or sickness funds collect contributions, pool risk, and reimburse health-care providers. Providers can be private, public, or nonprofit.

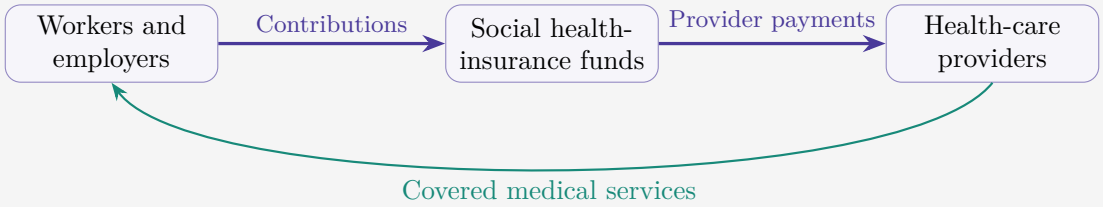


Figure 1.2: Bismarck financing: contributions are pooled to pay providers.

The key difference is the *principal source and institutional channel of financing*: general taxation versus organized social-insurance contributions. Real national systems combine features of both models.

1.2 The patient price, demand, and capacity

Question 1.3. Derive an individual's optimal demand for medical care when care has a positive out-of-pocket price.

Suppose a patient chooses consumption of ordinary goods and the quantity of health care. Ordinary consumption provides utility, and health-care use provides additional utility. Every purchased medical service reduces the budget available for ordinary goods. We assume both kinds of consumption are valuable and marginal benefit from health care diminishes. The optimum balances the extra utility from one more unit of care against the utility lost by forgoing ordinary consumption.

$$\begin{aligned}
\max_{c,h\geq 0} \quad & U(c,h) = u(c) + v(h), \\
\text{subject to} \quad & c + ph = y. \\
\text{Budget rearrangement:} \quad & c = y - ph. \\
\text{Substitution:} \quad & U(h) = u(y - ph) + v(h). \\
\text{First derivative:} \quad & \frac{dU}{dh} = u'(y - ph)(-p) + v'(h). \\
\text{Interior first-order condition:} \quad & -pu'(c^*) + v'(h^*) = 0. \\
& v'(h^*) = pu'(c^*).
\end{aligned}$$

Here the patient's out-of-pocket price is the opportunity cost of one additional treatment in units of ordinary consumption. When the price falls, that sacrificed consumption becomes smaller. The patient will generally desire more care, subject to preferences, medical relevance, and any remaining constraints. The equation is an interior condition; corner choices require checking feasible boundaries.

Question 1.4. Explain why zero patient prices can create excess demand and waiting times in a capacity-constrained health-care system.

Step 1: Demand effect. A zero monetary price removes the patient's direct payment for a covered service. If using additional care still provides a positive marginal private benefit, patients can request more services than they would request when paying a positive price.
Step 2: Capacity constraint. Hospitals still have finite staff, beds, operating budgets

and equipment. Thus the quantity patients desire can exceed the number of services the system can actually deliver.
Step 3: Allocation. When demand exceeds capacity, the extra requests cannot immediately be satisfied. Appointments and operations may be postponed, so waiting lists develop. Clinical priority, referrals and treatment guidelines may also allocate scarce slots.
Step 4: Nonmonetary price. Waiting imposes costs of time, inconvenience, uncertainty and sometimes deterioration in health. It therefore acts as an implicit price even when the cash price is zero. Free care at the point of use does not mean free production or unlimited access.

$$\begin{aligned} H_D(p) &= \text{aggregate quantity requested at patient price } p, \\ \bar{H} &= \text{maximum service capacity}, \\ H_D(0) &> \bar{H}, \\ ED &= H_D(0) - \bar{H} > 0. \end{aligned}$$

If patients value care sufficiently strongly at zero price, the number of requests exceeds available appointments or treatments. Excess demand is a count of unfilled requests, not additional medical capacity. The system must use waiting, priority, restrictions or investment to manage the gap.

Question 1.5. Derive why waiting time can function as an implicit price.

Let waiting time increase the nonmonetary cost of care, and suppose demand falls as that cost rises. Because the cost of waiting increases with the length of the wait, longer queues discourage some marginal demand. In a simplified model, an equilibrium waiting time is the wait at which requests equal the finite supply of appointments or treatments.

$$\begin{aligned} C'(w) > 0, \qquad H_D = H_D(C(w)), \qquad \frac{\partial H_D}{\partial C} < 0, \\ \frac{dH_D}{dw} = \frac{\partial H_D}{\partial C}\frac{dC(w)}{dw} < 0, \\ H_D(C(w^*)) = \bar{H}. \end{aligned}$$

The first derivative factor is negative because people request less care when the effective cost rises. The second is positive because longer waits are more costly. Their product is therefore negative. The equality defining equilibrium presumes suitable continuity, monotonicity and an attainable capacity level; actual queues can also depend on urgency, scheduling rules and dynamic backlogs.

1.3 The efficient amount of care

Question 1.6. Derive the efficient quantity of health care and verify that it maximizes social welfare.

Efficiency compares the marginal social benefit of an additional service with its real marginal resource cost. The illustrative lecture example assumes that the benefit of the next treatment declines as the total number of treatments rises, while the marginal resource

cost is constant. Total benefit is obtained by integrating marginal benefit; total cost is obtained by integrating marginal cost. The efficient service quantity maximizes their difference.

$$
\begin{aligned}
MB(h) &= 10 - h, & MC(h) &= 4.\\
MB(h^*) &= MC(h^*),\\
10 - h^* &= 4,\\
h^* &= 6.
\end{aligned}
$$

$$
TB(h) = \int_0^h (10 - x)\,dx = \left[10x - \frac{x^2}{2}\right]_0^h = 10h - \frac{h^2}{2},
$$

$$
TC(h) = \int_0^h 4\,dx = 4h,
$$

$$
\begin{aligned}
W(h) &= TB(h) - TC(h) = 6h - \frac{h^2}{2},\\
W'(h) &= 6 - h, & W'(h^*) = 0 &\implies h^* = 6,\\
W''(h) &= -1 < 0.
\end{aligned}
$$

The negative second derivative shows that social surplus is strictly concave and the critical quantity is a maximum. If zero-price private use continued until the private marginal benefit became zero, the illustrative quantity demanded would be ten rather than six. This particular difference is a model illustration, not a general claim that all additional insured care is unnecessary.

1.4 Social insurance and risk aversion

Question 1.7. Given a risk-averse individual with income Y, probability of illness q, and medical loss L, derive expected utility with and without insurance and explain why individuals demand health insurance in a Bismarck-type system.

Definitions. Let Y denote income before medical expenses, q the probability of illness, and L the medical loss if illness occurs. Thus $Y > L > 0$ and $0 < q < 1$. A healthy person does not suffer the medical loss; a sick person does. Risk aversion means that utility increases with consumption but at a decreasing rate: the loss of consumption in a bad state is especially painful.

Without insurance. Healthy-state consumption is the full income; sick-state consumption equals income less the medical loss. Expected utility is the probability-weighted average of utility in those two states.

$$
\begin{aligned}
&Y > L > 0, \qquad 0 < q < 1, \qquad u'(C) > 0, \quad u''(C) < 0,\\
&C_H^{(0)} = Y, \qquad C_S^{(0)} = Y - L,\\
&EU_0 = (1 - q)u(Y) + q\,u(Y - L).
\end{aligned}
$$

With insurance. Let the insured individual pay a premium and receive an indemnity if illness occurs. The premium reduces consumption in both states, while the indemnity partly or fully replaces income lost in the sick state. Expected utility again weights healthy-state and sick-state utility by their respective probabilities. Insurance will be

preferred when the smoothing benefit outweighs the utility cost of the premium.

$$
\begin{aligned}
P \geq 0 \quad &\text{(premium)},\\
0 \leq J \leq L \quad &\text{(indemnity)},\\
C_H^{(J)} &= Y - P,\\
C_S^{(J)} &= Y - P - L + J,\\
EU_J &= (1-q)u(Y-P) + q\,u(Y-P-L+J).\\
&\text{Insurance is preferred when } EU_J > EU_0.
\end{aligned}
$$

Economic explanation. A risk-averse person values reduced variation in consumption across healthy and sick states. By collecting contributions from a broad membership, a Bismarck-type insurance fund pools the financial risks of illness. Members contribute even when healthy; the fund pays eligible medical claims when some members become sick. Risk pooling makes individual medical payments less unpredictable but does not eliminate the medical resources required to treat illness. The desirability of a particular contract also depends on the contribution or premium and benefits offered.

Why risk aversion makes insurance valuable: a simple full-coverage comparison. To isolate the benefit of smoothing risk, consider the illustrative case in which the contribution equals the expected medical claim, $P = qL$, and the insured individual receives $J = L$ when sick. The insured individual then consumes the same amount in both states. For a concave utility function, Jensen's inequality shows that this certainty has at least as much utility as uncertain income with the same expected consumption.

$$
\begin{aligned}
C_H^{(J)} &= Y - qL,\\
C_S^{(J)} &= Y - qL - L + L = Y - qL,\\
EU_J &= u(Y - qL),\\
EU_0 &= (1-q)u(Y) + q\,u(Y-L),\\
EU_0 &\leq u\big((1-q)Y + q(Y-L)\big)\\
&= u(Y-qL) = EU_J.
\end{aligned}
$$

The inequality is strict when utility is strictly concave and both illness states occur with positive probability. This establishes the consumption-smoothing benefit of insurance in the benchmark. Actual Bismarck-type contributions need not equal this benchmark: compulsory participation and redistribution can change how costs are shared among members.

Question 1.8. Explain consumption smoothing and derive an illustrative full-coverage benchmark.

The simplest benchmark assumes the insurance pool receives the expected cost of claims and ignores loading and administration. Full coverage then exactly compensates the sick-state loss. Because healthy and sick consumption become equal, the insured patient no longer faces a consumption shortfall solely because illness occurred. This is a theoretical teaching benchmark; actual social insurance uses policy-determined contributions and

benefits.

$$\begin{aligned} P &= qJ, \qquad J = L, \qquad P = qL, \\ C_H &= Y - qL, \\ C_S &= Y - qL - L + L = Y - qL, \\ C_H &= C_S = Y - qL. \end{aligned}$$

With strictly concave and state-independent consumption utility, moving consumption from the healthy state toward the sick state can raise expected utility. The special equality above demonstrates complete smoothing for the hypothetical contract. Social-insurance contribution rates, subsidies, administration and coverage limits may prevent this equality in real systems.

Question 1.9. Explain risk pooling, adverse selection, and moral hazard in social health insurance.

Risk pooling combines the uncertain medical expenses of many enrollees. Broad participation can help distribute the burden of illness and protect a fund from extreme risk concentration. **Adverse selection** occurs when people know more about their likely medical costs than insurers and choose enrollment or plan generosity accordingly. If high-risk individuals enroll disproportionately, average claims increase, contributions or premiums may rise, and some low-risk people may leave. Compulsory participation, risk adjustment and subsidies are possible responses.
Ex post moral hazard is the change in use that may occur after insurance reduces the patient's out-of-pocket price. An increase in use is not automatically inefficient: some additional treatments may offer substantial health benefits. Good insurance policy balances financial protection against the opportunity cost of finite medical resources.

$$\begin{aligned} E[M] &= \sum_{i=1}^{N} q_i L_i, \\ p_c &= \alpha p, \qquad 0 \le \alpha \le 1, \\ \frac{\partial H_D}{\partial p_c} &< 0, \qquad \alpha \downarrow \implies p_c \downarrow \implies H_D \uparrow . \end{aligned}$$

The first expression sums each person's illness probability times the loss that would arise in that state. The second makes the distinction between the real resource price and the patient's share: more generous insurance generally lowers the latter. Changes in provider payments, treatment quality or eligibility can modify the size of the response.

1.5 Population aging and demographic indicators

Question 1.10. Define the elderly population share and the old-age dependency ratio.

The **elderly population share** is the fraction of the total population belonging to the elderly group. It measures how strongly older persons are represented in the population.

The **old-age dependency ratio** divides the elderly population by the number of working-age contributors. It measures the number of elderly people relative to the population potentially contributing labor income or social-insurance revenue. The younger or nonelderly population need not be identical to the actual contributor population, because children and some adults may not contribute.

$$N = N_Y + N_E, \qquad s_E = \frac{N_E}{N},$$
$$D = \frac{N_E}{N_W},$$

where N_Y is the younger population, N_E the elderly population and N_W the working-age contributing population.

Question 1.11. Under what additional assumption can the old-age dependency ratio be expressed using only the elderly share?

Assume, solely for this simplified two-group example, that everyone who is not elderly is a working-age contributor. Then the size of the contributor group equals the total population minus the elderly population. Under that assumption only, the dependency ratio can be expressed entirely in terms of the elderly share.

$$N_E = s_E N,$$
$$N_W = N - N_E = N - s_E N = N(1 - s_E),$$
$$D = \frac{N_E}{N_W} = \frac{s_E N}{N(1 - s_E)} = \frac{s_E}{1 - s_E},$$
$$\frac{dD}{ds_E} = \frac{1}{(1 - s_E)^2} > 0 \qquad (0 \le s_E < 1).$$

This derivative is positive: within the simplified model, a higher elderly share also increases the number of elderly people per contributor. The formula is not generally valid when the nonelderly population includes children or people who are not working-age contributors.

Question 1.12. Given total population N, per-person health expenditures e_Y and e_E, and elderly population share s_E, derive total health expenditure as a function of s_E. Show the effect of aging when $e_E > e_Y$.

Let per-person expenditure for the younger group be represented by one parameter, and per-person expenditure for the elderly group by another. Multiply each group's per-person expense by the number of people in that group, then add both groups. To express expenditure using the elderly share, replace the number of elderly people by the share times total population and replace the younger population by the remaining share times total population. Finally, collect the terms that contain the elderly share.

$$
\begin{aligned}
E &= e_Y N_Y + e_E N_E, \\
N_E &= s_E N, \\
N_Y &= N - N_E = N - s_E N = (1 - s_E)N, \\
E &= e_Y(1 - s_E)N + e_E s_E N, \\
&= N\big[e_Y - e_Y s_E + e_E s_E\big], \\
E(s_E) &= N\big[e_Y + (e_E - e_Y)s_E\big]. \\
\frac{\partial E}{\partial s_E} &= N\frac{\partial}{\partial s_E}\big[e_Y + (e_E - e_Y)s_E\big], \\
&= N\big[0 + (e_E - e_Y)\cdot 1\big], \\
\frac{\partial E}{\partial s_E} &= N(e_E - e_Y) > 0 \quad \text{if } e_E > e_Y.
\end{aligned}
$$

The derivative holds total population and the two per-person spending levels fixed. Because the elderly group has higher per-person spending in the stated case, shifting a larger proportion of a fixed population into that group raises total spending. This is a *composition effect*. It does not imply that aging is the only cause of expenditure growth, nor does it establish that older individuals always have the same spending needs in practice.

1.6 Health-system budgets and financial sustainability

Question 1.13. What is a health-system financing gap?

A financing gap is expenditure minus the revenue assigned to health care during the same period. A positive gap denotes a current deficit, zero denotes current balance, and a negative gap a current surplus. A balanced flow in one year is not sufficient to prove long-run sustainability because demographics, medical costs, tax receipts, reserves and economic growth may change.

$$
\begin{aligned}
&G_t = E_t - R_t, \\
&G_t > 0 \text{ deficit}, \qquad G_t = 0 \text{ balance}, \qquad G_t < 0 \text{ surplus}.
\end{aligned}
$$

Question 1.14. Analyze how population aging affects the financial sustainability of Beveridge-type health-care systems.

First, spending pressure. Beveridge-type systems finance care mainly through taxation. If average elderly health spending exceeds average younger-person spending, a higher elderly share raises aggregate expected medical expenditure at a fixed population size.
Second, fiscal pressure. If the health budget's assigned tax revenue does not increase at the same pace, the financing gap grows. Depending on employment, income and taxation, a larger retired population may also weaken revenue growth relative to health needs. That extra effect is not automatic and should be analyzed separately.
Third, capacity pressure. If older persons use more care on average while hospitals remain capacity-constrained, demand may exceed available services by a greater amount. This can lengthen waiting times and increase the need for clinical priority rules.

Fourth, policy choices. Governments can adjust tax allocations, expand the revenue base, improve preventive care, increase productivity, invest in capacity, and improve treatment selection. Thus aging creates a potential sustainability challenge, not an inevitable collapse of the system.

$$\begin{aligned} G(s_E) &= E(s_E) - R \\ &= N\big[e_Y + (e_E - e_Y)s_E\big] - R, \\ \frac{\partial G}{\partial s_E} &= \frac{\partial E}{\partial s_E} - \frac{\partial R}{\partial s_E}, \\ \frac{\partial R}{\partial s_E} &= 0 \quad \text{under the specified fixed-revenue comparison,} \\ \frac{\partial G}{\partial s_E} &= N(e_E - e_Y) > 0 \quad \text{if } e_E > e_Y. \end{aligned}$$

This comparison isolates age composition. It does not hold if tax revenue, population size, health-care prices or group-specific medical spending also respond to aging. In a broader model, total spending and revenue may change together; the sign of the net fiscal effect must then be calculated from their combined changes.

Question 1.15. Derive how aging may increase aggregate health-care demand and waiting times.

Suppose an average elderly person uses more services than an average younger person. Aggregate desired use equals use per person times the number of people in each group. Substituting the elderly share shows that aging shifts aggregate demand upward. If treatment capacity stays fixed, more desired services create additional potential excess demand.

$$\begin{aligned} H_D &= h_Y N_Y + h_E N_E, \\ &= N\big[h_Y + (h_E - h_Y)s_E\big], \\ \frac{\partial H_D}{\partial s_E} &= N(h_E - h_Y) > 0 \quad \text{if } h_E > h_Y, \\ ED &= H_D - \bar{H}, \\ \frac{\partial ED}{\partial s_E} &= N(h_E - h_Y) > 0 \quad \text{if capacity is fixed.} \end{aligned}$$

To explain how waiting time adjusts, treat desired services as decreasing in waiting cost and increasing in the elderly share. Equilibrium means demand equals fixed capacity. Differentiate that equality with respect to the elderly share. The multivariable chain rule captures both the direct increase in demand from aging and the indirect reduction in

demand caused by a longer wait.

$$
\begin{aligned}
H_D(C(w^*(s_E)), s_E) &= \bar{H}, \\
\frac{d}{ds_E} H_D(C(w^*(s_E)), s_E) &= \frac{d\bar{H}}{ds_E} = 0, \\
\frac{\partial H_D}{\partial C}\frac{dC}{ds_E} + \frac{\partial H_D}{\partial s_E} &= 0, \\
\frac{dC}{ds_E} &= C'(w^*)\frac{dw^*}{ds_E}, \\
\frac{\partial H_D}{\partial C} C'(w^*)\frac{dw^*}{ds_E} + \frac{\partial H_D}{\partial s_E} &= 0, \\
\frac{\partial H_D}{\partial C} C'(w^*)\frac{dw^*}{ds_E} &= -\frac{\partial H_D}{\partial s_E}, \\
\frac{dw^*}{ds_E} &= -\frac{\partial H_D/\partial s_E}{(\partial H_D/\partial C)C'(w^*)} > 0.
\end{aligned}
$$

The numerator is positive because aging raises desired use; the demand derivative with respect to waiting cost is negative; and longer waits increase that cost. The denominator is negative, so the expression is positive. This predicted rise in equilibrium wait is conditional on fixed capacity, an interior solution and the model's monotonicity assumptions.

1.7 Aging and Bismarck insurance contributions

Question 1.16. Derive the balanced-budget social-insurance contribution rate.

In a Bismarck-type system, revenue can be modeled as the contribution rate multiplied by average contribution-eligible earnings and the number of contributors. When the insurance fund's revenue exactly equals medical expenditure, algebra gives the contribution rate needed to finance those benefits. Higher spending requires a higher rate if earnings and contributors remain fixed, whereas a larger contribution base reduces the required rate.

$$
\begin{aligned}
R &= \tau \bar{y} N_W, \\
R = E &\implies \tau^* \bar{y} N_W = E, \\
\tau^* &= \frac{E}{\bar{y} N_W} \qquad (\bar{y} N_W > 0), \\
\tau^* &= \frac{e_Y N_Y + e_E N_E}{\bar{y} N_W}, \\
\frac{\partial \tau^*}{\partial N_E}\Big|_{N_Y, N_W, \bar{y}, e_Y, e_E} &= \frac{e_E}{\bar{y} N_W} > 0.
\end{aligned}
$$

The final partial derivative deliberately holds all other quantities fixed; it captures only the added expense associated with more elderly beneficiaries. In practice aging may also alter the number of contributors or their taxable earnings, potentially increasing the revenue pressure.

Question 1.17. Show the special relationship between the old-age dependency ratio and the contribution rate.

For a particularly simple teaching example, ignore younger-group medical spending and suppose elderly spending dominates the fund. Total medical expenditure then equals elderly per-person cost times the elderly population. Replacing elderly persons per contributor by the dependency ratio gives a direct relationship between that ratio and the rate needed for budget balance.

$$\begin{aligned} E &= e_E N_E, \\ \tau^* \bar{y} N_W &= e_E N_E, \\ \tau^* &= \frac{e_E N_E}{\bar{y} N_W} \\ &= \frac{e_E}{\bar{y}} \frac{N_E}{N_W}, \\ \tau^* &= \frac{e_E}{\bar{y}} D, \qquad \frac{\partial \tau^*}{\partial D} = \frac{e_E}{\bar{y}} > 0. \end{aligned}$$

This formula applies only to the stated special case; the full model must include younger-group care, actual contributors and all revenue sources. It makes clear why aging can challenge earnings-based insurance financing when beneficiary spending rises faster than the contribution base.

Question 1.18. Compare the main sustainability mechanisms in Beveridge and Bismarck systems.

Both systems must finance higher medical spending when a population requires more treatment. A Beveridge-type system primarily faces a tension between tax-financed budgets and the capacity to deliver services; a Bismarck-type system primarily faces a tension between insured benefits and revenues based on contributions. Both can experience increased demand from aging and more intense utilization when point-of-use prices are low.

Neither model can avoid scarcity. Policy choices include prevention, better procurement, productivity improvements, contributor-base or tax-base reforms, and carefully targeted capacity investments. An aging population can heighten pressure in both models, but its net effect depends on employment, taxes, technology, health status and policy reform.

CHAPTER 2

Economic Epidemiology and the SIR Model

Learning goals. Identify the three disease states; derive the mass-action infection and recovery flows; construct and interpret the differential equations; calculate reproduction numbers and the peak condition; analyze prevention, transmission externalities and hospital congestion.

2.1 Why economists study infection dynamics

Question 2.1. Why does infectious disease have an economic dimension?

An individual's health risk depends partly on other people's disease status and interactions. Decisions about work, transport, school, prevention and contact can expose others to infection. Illness also changes demand for medical services, labor supply and household expenditure. The SIR model provides a simplified, transparent description of the epidemic process: an infectious outbreak can grow, peak and decline as new infections and recoveries change over time.

2.2 SIR compartments and the closed-population assumption

Question 2.2. Define the susceptible, infectious, and recovered populations in the SIR model.

Susceptible: People who are not currently infected but can acquire the disease upon effective exposure. They leave this compartment when newly infected.
Infectious: People currently capable of transmitting the disease to others. They enter through infection and leave when they recover or are otherwise removed from infectiousness.
Recovered: People who have left the infectious state and, in the basic model, remain immune rather than becoming susceptible again.
The basic model assumes a closed population with no births, deaths, reinfections or demographic change during the studied period. Everyone belongs to exactly one of the

three compartments.

$$
\begin{aligned}
S(t) &= \text{susceptible population at time } t,\\
I(t) &= \text{infectious population at time } t,\\
R(t) &= \text{recovered population at time } t,\\
N &= S(t) + I(t) + R(t), \quad N \text{ constant.}
\end{aligned}
$$

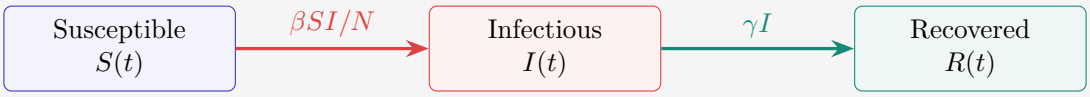


Figure 2.1: Susceptible–infectious–recovered population flows.

2.3 Derivation of infection and recovery flows

Question 2.3. Why is the new-infection flow proportional to the susceptible population and the infectious share?

Suppose each susceptible person has a given number of potentially infectious contacts per day. With random mixing, the chance a contact is with someone who is infectious is approximately the infectious share of the whole population. Conditional on meeting an infectious person, transmission occurs with a given probability. Multiply contacts, the infectious share and transmission probability to obtain expected new infections per susceptible person; then multiply by the number of susceptible people.

$$
\begin{aligned}
c &= \text{contacts per person per unit time},\\
p &= \text{transmission probability per infectious contact},\\
\Pr(\text{infectious contact}) &= \frac{I}{N},\\
f_{\text{person}} &= cp\frac{I}{N},\\
F_{\text{new}} &= S\, f_{\text{person}} = S\, cp\frac{I}{N},\\
\beta &= cp,\\
F_{\text{new}} &= \beta\frac{SI}{N}.
\end{aligned}
$$

The parameter for transmission combines contact frequency and transmission risk per contact. The formula assumes homogeneous random mixing: an infectious person is sampled in proportion to the infectious share. It is a model of expected flows, not a claim that any given person makes fractional contacts or that real populations mix perfectly.

Question 2.4. Derive the recovery flow and interpret the recovery parameter.

Assume each currently infectious person has a constant chance of recovery per short unit of time, independently of the number susceptible. During a short interval, the expected number of recoveries is the per-person recovery rate multiplied by the infectious population and by interval length. Dividing by the interval length gives the recovery flow per unit

time.

$$\gamma > 0 = \text{constant recovery/removal hazard},$$
$$\Pr(\text{recovery in } \Delta t) \approx \gamma \Delta t,$$
$$\text{expected recoveries in } \Delta t \approx I(t)\gamma\Delta t,$$
$$\text{recoveries per unit time} \approx \frac{I(t)\gamma\Delta t}{\Delta t},$$
$$\text{recovery flow} = \gamma I(t), \qquad \mathbb{E}[\text{infectious duration}] = \frac{1}{\gamma}.$$

Under a constant hazard, the average infectious duration equals the reciprocal of the recovery rate. For example, a daily recovery rate of one tenth implies a mean infectious period of ten days. Actual infectious periods vary among individuals and diseases.

2.4 Constructing the three differential equations

Question 2.5. Given new infections $\beta SI/N$ and recoveries γI, derive the three differential equations of the SIR model for a closed population.

Step 1: Susceptibles. Every new infection removes one person from the susceptible group. Hence the change in susceptible people is negative and equal in magnitude to the new-infection flow.
Step 2: Infectious people. Every new infection adds one infectious person, while every recovery removes one. The net change is new infections minus recoveries.
Step 3: Recovered people. Recoveries add people to the recovered group. Under the basic assumptions, there is no flow out of that group.
Step 4: Continuous time. Write the changes over a short interval, divide by the interval length, and take the limit as the interval approaches zero.

$$S(t+\Delta t) - S(t) \approx -\beta \frac{S(t)I(t)}{N}\Delta t,$$
$$I(t+\Delta t) - I(t) \approx \left[\beta \frac{S(t)I(t)}{N} - \gamma I(t)\right]\Delta t,$$
$$R(t+\Delta t) - R(t) \approx \gamma I(t)\Delta t.$$

$$\lim_{\Delta t \to 0} \frac{S(t+\Delta t) - S(t)}{\Delta t} = -\beta \frac{S(t)I(t)}{N},$$
$$\lim_{\Delta t \to 0} \frac{I(t+\Delta t) - I(t)}{\Delta t} = \beta \frac{S(t)I(t)}{N} - \gamma I(t),$$
$$\lim_{\Delta t \to 0} \frac{R(t+\Delta t) - R(t)}{\Delta t} = \gamma I(t),$$

$$\frac{dS}{dt} = -\beta \frac{SI}{N}, \qquad \frac{dI}{dt} = \beta \frac{SI}{N} - \gamma I, \qquad \frac{dR}{dt} = \gamma I.$$

The signs match the movements shown in the diagram. The susceptible stock falls when infection occurs. The infectious stock can either rise or fall depending on whether the new-infection flow exceeds the recovery flow. The recovered stock never falls in the basic model. These equations concern changes in population *stocks*; the infection and recovery expressions are *flows* per unit time.

Question 2.6. Prove that the three SIR equations conserve the population.

The model assumes that no people enter or leave the population. We can verify the assumption directly by adding the three rates of change. Every person who leaves the susceptible compartment appears in the infectious compartment. Likewise, every recovery moves one person out of the infectious compartment and into the recovered compartment. Therefore each transfer cancels in the population total.

$$\begin{aligned} N &= S + I + R, \\ \frac{dN}{dt} &= \frac{dS}{dt} + \frac{dI}{dt} + \frac{dR}{dt}, \\ &= -\beta\frac{SI}{N} + \left(\beta\frac{SI}{N} - \gamma I\right) + \gamma I, \\ &= \left(-\beta\frac{SI}{N} + \beta\frac{SI}{N}\right) + (-\gamma I + \gamma I), \\ \frac{dN}{dt} &= 0. \end{aligned}$$

Because the derivative is zero, the total population remains constant over time under these assumptions. Models with births, deaths, migration or loss of immunity need additional terms, and the population identity may change accordingly.

2.5 Reproduction numbers, early growth, and epidemic peaks

Question 2.7. Derive the basic reproduction number and the condition for an interior peak in the infectious population. Explain how reducing transmission contributes to disease control.

Basic reproduction number. At the start of an epidemic, suppose almost everyone is susceptible. One infectious person produces new infections at approximately the transmission rate per unit time and remains infectious for the mean duration determined by the recovery hazard. Multiplying those two quantities gives the expected number of secondary infections in an otherwise susceptible population.

$$\begin{aligned} \text{infection rate per infectious person} &\approx \beta \quad \text{when } S \approx N, \\ \text{mean infectious duration} &= \frac{1}{\gamma}, \\ R_0 &= \beta\left(\frac{1}{\gamma}\right) = \frac{\beta}{\gamma}. \end{aligned}$$

If the basic reproduction number exceeds one, a small introduction into an almost fully susceptible population can initially grow; below one, it tends to decline. This threshold is based on the simplified homogeneous-mixing model, not a universal prediction for every real outbreak.

Peak condition. At an interior peak of the infectious population, new infections exactly balance recoveries. To derive the condition, set the time derivative of the infectious stock equal to zero. Provided the number infectious is positive, divide through by that number

and rearrange.

$$\begin{aligned}
\frac{dI}{dt} &= \beta\frac{SI}{N} - \gamma I, \\
\left.\frac{dI}{dt}\right|_{t=t_{\text{peak}}} &= 0, \\
0 &= \beta\frac{S_{\text{peak}}I_{\text{peak}}}{N} - \gamma I_{\text{peak}}, \\
0 &= I_{\text{peak}}\left(\frac{\beta S_{\text{peak}}}{N} - \gamma\right), \\
0 &= \frac{\beta S_{\text{peak}}}{N} - \gamma \quad (I_{\text{peak}} > 0), \\
\frac{\beta S_{\text{peak}}}{N} &= \gamma, \\
\frac{S_{\text{peak}}}{N} &= \frac{\gamma}{\beta} = \frac{1}{R_0}.
\end{aligned}$$

The peak occurs when each existing infectious case just replaces itself on average. Before that condition, the infectious stock may rise; once the effective reproduction number falls below one, it declines. The peak does *not* mean there are no further infections; it means the inflow and outflow of infectious people are equal at that instant.

Disease control. Fewer potentially infectious contacts or a lower transmission probability per contact reduces the transmission parameter. Holding the other variables fixed, that reduces the inflow of newly infected people and lowers the reproduction numbers. In the lecture's comparison, lower transmission produces a lower and later infectious peak, helping to protect scarce hospital capacity. Actual effects depend on the timing and duration of prevention, population behavior and the disease's characteristics.

$$\begin{aligned}
R_t &= \frac{\beta}{\gamma}\frac{S(t)}{N} = R_0\frac{S(t)}{N}, \\
\frac{dI}{dt} &= \beta\frac{SI}{N} - \gamma I \\
&= \gamma I\left(\frac{\beta S}{\gamma N} - 1\right) \\
&= \gamma I(R_t - 1), \\
R_t > 1 &\implies \frac{dI}{dt} > 0 \quad (I > 0), \\
R_t = 1 &\implies \frac{dI}{dt} = 0, \\
R_t < 1 &\implies \frac{dI}{dt} < 0 \quad (I > 0), \\
\beta &= cp.
\end{aligned}$$

This final identity shows two levers for prevention: reduce risky contact frequency or lower infection probability when contact occurs. The qualitative mechanism follows directly from the equations; interventions also have economic and social costs that must be considered when selecting public-health policy.

Question 2.8. Derive and interpret the simple herd-immunity threshold.

In the basic model, if recovered people retain immunity and the non-susceptible fraction is immune, the threshold at which the effective reproduction number reaches one can be converted into an immune fraction. This is a theoretical threshold under homogeneous mixing and constant parameters; it is not by itself a vaccination target for a real country or disease.

$$
\begin{aligned}
1 = R_t &= R_0 \frac{S}{N}, \\
\frac{S}{N} &= \frac{1}{R_0}, \\
H_c &= 1 - \frac{S}{N}, \\
H_c &= 1 - \frac{1}{R_0}, \qquad (R_0 > 1), \\
R_0 = 3 \implies H_c &= 1 - \frac{1}{3} = \frac{2}{3} \approx 66.7\%.
\end{aligned}
$$

Crossing this threshold starts a decline in active infectious cases in the simple model. It does not stop infections instantaneously: existing infectious people can continue to transmit, and the eventual immune fraction may exceed the threshold. Heterogeneous contacts, changing immunity and new variants can change real-world thresholds.

2.6 Reading epidemic curves and calculating an example

Question 2.9. How should the susceptible, infectious, and recovered curves be interpreted?

The susceptible curve falls because susceptibles leave through infection and are not replenished in this simple model. The recovered curve rises because recoveries accumulate. The infectious curve often rises early and then falls: at first many people are susceptible, permitting new infections to outnumber recoveries; later the shrinking susceptible pool makes recovery outflow exceed new-infection inflow. The enclosed graph reproduces the lecture's illustrative parameter values through a numerical solution of the SIR equations.

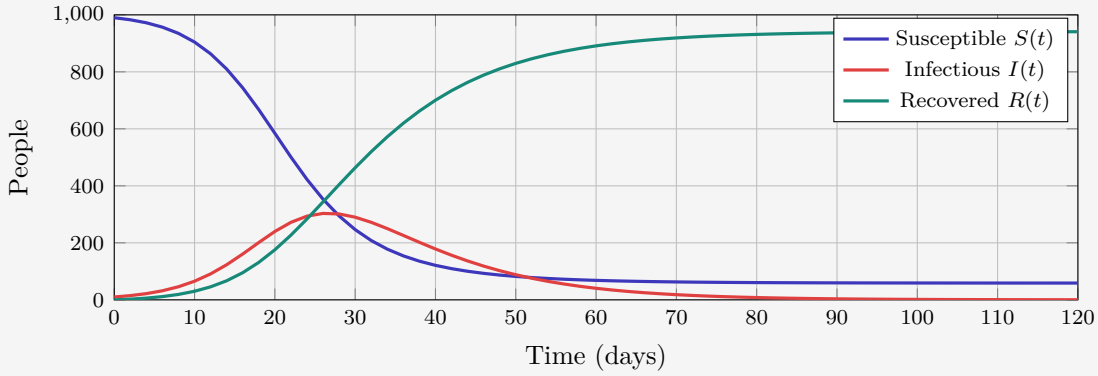


Figure 2.2: SIR dynamics with $N = 1000$, $S(0) = 990$, $I(0) = 10$, $\beta = 0.30$, $\gamma = 0.10$ per day.

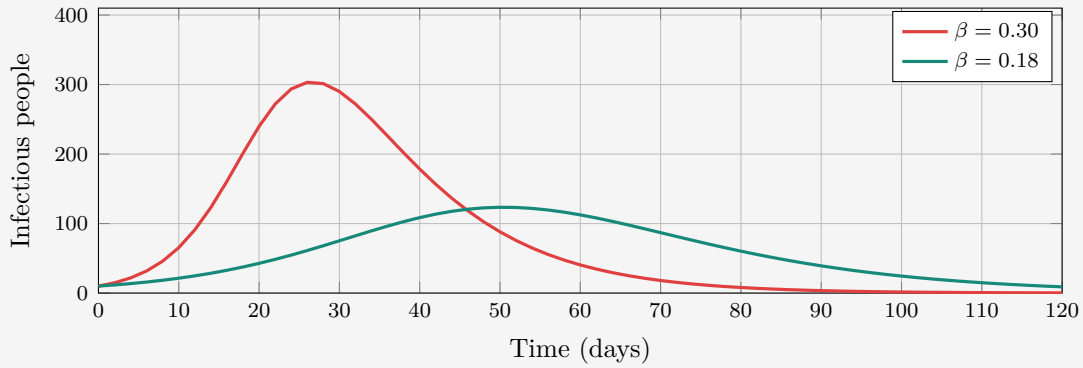


Figure 2.3: Lower transmission lowers and delays the infection peak in the illustrated model.

Question 2.10. Calculate all three instantaneous flows and both reproduction numbers for the lecture's numerical example.

The example begins with one thousand people, of whom ten are infectious and the rest susceptible. There are five potentially risky contacts per person per day; the chance of infection from a contact with an infectious person is six percent, and the daily recovery hazard is one tenth. First combine contact rate and probability to obtain transmission. Next multiply transmission by the susceptible–infectious product divided by population to obtain new infections. Finally compute recoveries and subtract them from infections to find the change in the infectious population.

$$
\begin{aligned}
N = 1000, \quad S(0) = 990, \quad I(0) = 10, \quad R(0) = 0,& \\
c = 5, \qquad p = 0.06, \qquad \gamma = 0.10 \text{ day}^{-1},& \\
\beta = cp = 5(0.06) = 0.30 \text{ day}^{-1},& \\
\text{new infections/day} = 0.30\frac{990(10)}{1000} = 2.97,& \\
\text{recoveries/day} = 0.10(10) = 1.00,& \\
\left.\frac{dS}{dt}\right|_0 = -2.97,& \\
\left.\frac{dI}{dt}\right|_0 = 2.97 - 1.00 = 1.97,& \\
\left.\frac{dR}{dt}\right|_0 = 1.00,& \\
R_0 = \frac{0.30}{0.10} = 3,& \\
R_{t=0} = 3\frac{990}{1000} = 2.97.&
\end{aligned}
$$

At the starting instant, the infectious stock is increasing by approximately 1.97 persons per day in the continuous model. The noninteger flows are expected rates, not literal counts of fractional individuals. The stock-conservation check also works: the three rates sum to zero.

2.7 Prevention and infectious-disease externalities

Question 2.11. Explain why private decisions can lead to too little infectious-disease prevention.

A person may take precautions because infection could harm their own health or income. But the same prevention can reduce transmission to classmates, coworkers, family members and strangers. These benefits to others are positive externalities. When a person considers only the private benefit, the chosen level of prevention may be below the socially desirable level. Public-health information, vaccination support and other proportionate measures can help internalize this external benefit.

$$\begin{aligned} MSB(a) &= MPB(a) + MEB(a), \\ MPB(a_{\text{private}}) &= MC(a_{\text{private}}), \\ MSB(a_{\text{social}}) &= MC(a_{\text{social}}). \end{aligned}$$

Here prevention effort is a choice variable; private marginal benefit describes a person's own avoided harm, external marginal benefit describes others' avoided harm, and marginal cost captures foregone time, income, convenience or public spending. Under standard diminishing-benefit assumptions and positive external marginal benefit, social optimal prevention tends to be greater than privately chosen prevention. These conditions matter: no single prevention policy is automatically optimal for every epidemic.

2.8 Hospital congestion and financing pressures

Question 2.12. Relate the infectious population to hospital capacity and health-system finance.

For a rough teaching approximation, suppose a fixed fraction of infectious people requires a hospital place. The number of required places is then proportional to the infectious stock. When required beds exceed available beds, patients may be delayed or rerouted, and clinical outcomes can be affected. A transmission-reducing intervention that lowers the infection peak can reduce maximum pressure on hospital capacity.

$$\begin{aligned} B(t) &= \kappa I(t), \\ B(t) > K &\iff \kappa I(t) > K, \\ I(t) &> \frac{K}{\kappa} \qquad (\kappa > 0). \end{aligned}$$

The simple proportionality is not a clinical forecast: real hospital occupancy reflects age, severity, delayed admissions, length of stay and available treatments. In a Beveridge system, a surge strains tax-funded budgets, medical capacity and waiting lists. In a Bismarck system, it can raise insured claims and reimbursements. Prevention can therefore create both health benefits and financial or capacity benefits.

Question 2.13. State the principal limitations of the basic SIR model.

The basic model assumes homogeneous random mixing, constant transmission and recovery parameters, an immediate infectious stage, no births or deaths, and lasting immunity after recovery. Real diseases exhibit variation in behavior, geographic networks, contact rates, clinical severity and biological immunity. SEIR models add an exposed but not yet infectious stage; other extensions allow time-varying transmission, vaccination, waning immunity, reinfection, age structure and hospital states. Quantitative policy recommendations require data-driven parameter estimates and a model suited to the clinical setting.

CHAPTER 3

Pharmaceutical Innovation

Learning goals. Explain the sequence of drug development; calculate probabilities and portfolio R&D costs; analyze patents, generic entry, monopoly pricing and induced research incentives; distinguish regulation, cost effectiveness and affordability; derive medical-expenditure growth.

3.1 Pharmaceutical research as an economic activity

Question 3.1. Why is pharmaceutical innovation economically different from ordinary production?

The seller of a new medicine must invest substantial resources before knowing whether the product will be medically effective, safe enough for use, approved by regulators, or commercially successful. Laboratory screening, preclinical research, clinical trials, scientific documentation and manufacturing development create large upfront costs. The direct cost of producing one additional pill may be much smaller than the total research cost. Because patients usually cannot independently verify safety or effectiveness, trustworthy evidence and regulation play important roles. Society simultaneously wants incentives for the next discovery and affordable access to existing treatments.

3.2 The life cycle of a pharmaceutical drug

Question 3.2. Identify and briefly describe the main stages in the life cycle of a pharmaceutical drug.

A medicine normally passes through **discovery and screening**, **preclinical research**, **clinical trials**, **regulatory review and approval**, and **market launch and patient access**. After approval, **postmarket safety monitoring and additional study** continue. When effective exclusive rights expire and relevant regulatory requirements are met, **generic or biosimilar competition** may enter. Candidates can fail or be abandoned at several stages, and real approval pathways need not follow an identical sequence for every product.

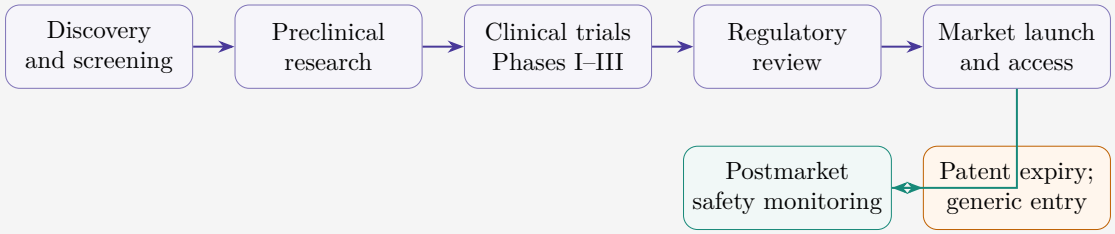


Figure 3.1: Drug development, regulatory review, and postmarket stages.

Question 3.3. What happens in discovery, screening, and preclinical research?

Discovery and screening. Researchers identify a disease mechanism or biological target, investigate potential compounds and look for candidates with useful biological activity. Promising candidates are optimized for effectiveness, stability and tolerability. Much of the output is new scientific information, not immediate revenue.
Preclinical research. Developers use laboratory work and, where required and appropriate, animal studies to assess how a compound acts, how it is processed by the body, and possible toxicities. This evidence helps decide whether potential benefits justify the risk of exposing human volunteers. Passing this stage does not guarantee safety or effectiveness in patients.

Question 3.4. Distinguish the main purposes of Phase I, Phase II and Phase III clinical trials.

Phase I focuses primarily on initial human safety, tolerability and the way the body handles the drug. **Phase II** asks whether there is promising activity or clinical benefit in the target patient population, while continuing safety evaluation. **Phase III** generally collects stronger evidence of effectiveness and safety in larger or more relevant groups to support a regulatory application. These phases are costly information-generating investments and can lead to discontinuation of an unsuccessful candidate. Trial size, design and sequencing vary by product and regulatory pathway.

Stage	Main scientific question	Economic significance
Phase I	Initial safety, tolerability and pharmacology	Reduce initial uncertainty in humans
Phase II	Signs of clinical activity and appropriate dosing	Decide whether larger studies are justified
Phase III	Substantial evidence of benefit and safety	Large investment before possible filing

Question 3.5. Distinguish regulatory approval, market access, and postmarket monitoring.

Approval means the responsible regulator judges that the submitted evidence satisfies the applicable benefit–risk and quality requirements for marketing. **Market access** concerns price, reimbursement, insurance coverage, provider adoption and whether eligible patients can actually obtain the medicine. **Postmarket monitoring** collects new evidence about rare adverse events, long-term effects and use in additional populations. Regulatory approval does not establish that a medicine is free from all risk, covered by every insurer,

cost-effective or affordable to all patients.

3.3 Why pharmaceutical development is costly and uncertain

Question 3.6. Explain how uncertainty, failed projects, and clinical trials contribute to the cost of developing a successful medicine.

Uncertainty. Research begins before firms know whether a substance will work, be safe, receive approval or attract enough users. Scientific, clinical, regulatory, commercial and competitive uncertainties all create a risk of failure.
Failed candidates. Expenditure on abandoned projects is part of the research portfolio that eventually produces successful drugs. When only a minority of candidates reach approval, the cost of the whole portfolio must be supported by fewer commercial successes.
Clinical trials. Phase I, II and III studies require trial participants, clinicians, monitoring, data management, analysis, manufacturing and compliance. Later, larger trials often cost much more per candidate. A failed late-stage candidate can therefore consume substantial resources without producing a saleable medicine.
Delay and capital cost. Drug development can take many years. Resources invested today have an opportunity cost because revenues, if any, arrive later. The cost of developing a successful medicine therefore reflects direct spending, unsuccessful trials, the passage of time and, in some cases, postapproval obligations. There is no single universal cost figure for all drugs.

Question 3.7. Using the lecture's illustrative portfolio, calculate total clinical-phase cost and cost per approved product.

The example begins with one hundred candidates in Phase I. Only sixty enter Phase II, twenty-four enter Phase III, and twelve ultimately receive approval. Each candidate that enters a phase incurs that phase's stipulated cost. To obtain portfolio expenditure, multiply entrants by cost per entrant in each phase and sum. Finally, divide total clinical-phase spending by approved products.

Phase	Candidates entering	Cost per entrant (million US$)	Total (million US$)
I	100	10	1,000
II	60	30	1,800
III	24	100	2,400
Total			5,200

$$
\begin{aligned}
C_{\mathrm{I}} &= 100(10) = 1{,}000 \text{ million US\$},\\
C_{\mathrm{II}} &= 60(30) = 1{,}800 \text{ million US\$},\\
C_{\mathrm{III}} &= 24(100) = 2{,}400 \text{ million US\$},\\
C_{\text{clinical}} &= 1{,}000 + 1{,}800 + 2{,}400\\
&= 5{,}200 \text{ million US\$},\\
\text{cost per approved drug} &= \frac{5{,}200}{12}\\
&\approx 433.33 \text{ million US\$}.
\end{aligned}
$$

The numerical values are deliberately hypothetical. This average includes the trial spending on unsuccessful candidates but excludes discovery, preclinical work, opportunity cost of capital and postapproval expenses. It must not be interpreted as an observed or universally applicable cost of developing a drug.

Question 3.8. Derive the overall probability of a candidate's success from conditional stage probabilities.

A drug must pass several gates in order to be approved. The probability of passing all gates is calculated by multiplying the probability of passing the first gate, the conditional probability of passing the second given first-stage success, and the conditional probability of passing the third given previous success. This is the chain rule of probability, not a requirement that stage outcomes be independent.

$$
\begin{aligned}
A_1 &= \text{passing stage 1}, \quad A_2 = \text{passing stage 2}, \quad A_3 = \text{final success},\\
\Pr(A_1 \cap A_2 \cap A_3) &= \Pr(A_1)\Pr(A_2 \mid A_1)\Pr(A_3 \mid A_1, A_2),\\
q_1 &= \Pr(A_1), \quad q_2 = \Pr(A_2 \mid A_1), \quad q_3 = \Pr(A_3 \mid A_1, A_2),\\
p_{\text{success}} &= q_1 q_2 q_3,\\
p_{\text{success}} &= (0.60)(0.40)(0.50) = 0.12,\\
\mathbb{E}[\text{approvals}] &= 100(0.12) = 12.
\end{aligned}
$$

In this teaching example, the expected number of approvals is twelve from one hundred starting candidates. The probabilities are illustrative and not estimates for a particular therapy, disease or year.

Question 3.9. Derive expected clinical R&D spending per successful product in a large portfolio.

Each candidate entering the portfolio incurs first-stage cost. Only candidates that clear the first gate incur second-stage cost, and only candidates that clear the first two gates incur third-stage cost. We therefore weight the later stage costs by the chance that a starting candidate reaches each stage. Divide expected cost per starting candidate by the probability that a starting candidate is eventually approved. This produces the

large-portfolio cost per success under homogeneous candidate probabilities and costs.

$$\begin{aligned} C &= c_1 + \mathbf{1}_{A_1} c_2 + \mathbf{1}_{A_1 \cap A_2} c_3, \\ \mathbb{E}[C] &= c_1 + q_1 c_2 + (q_1 q_2) c_3, \\ p_{\text{success}} &= q_1 q_2 q_3, \\ C_{\text{per success}} &= \frac{c_1 + q_1 c_2 + q_1 q_2 c_3}{q_1 q_2 q_3}. \end{aligned}$$

The numerator records spending incurred even by candidates that fail; the denominator counts expected successes. A smaller probability of success generally means more starting projects and more failed-project spending must be supported by each approved product, with other assumptions held fixed. This ratio describes expected portfolio spending per expected success, not the expected expenditure of a single project conditional on success.

Question 3.10. Why does the time value of money matter in pharmaceutical research?

Drug research and clinical trials require substantial investment years before revenue arrives. Money spent today could instead have funded other research or earned returns. The present value of a future receipt discounts it by the number of years before collection. Long delays reduce the present value of future earnings and can make an otherwise promising project financially unattractive.

$$\begin{aligned} PV(F) &= \frac{F}{(1+r)^T}, \\ F &= 100 \text{ million US\$}, \quad r = 0.05, \quad T = 10, \\ PV &= \frac{100}{(1.05)^{10}} \\ &\approx 61.39 \text{ million US\$}. \end{aligned}$$

One hundred million dollars received ten years later is worth about sixty-one million dollars today at the assumed five-percent annual rate. This comparison illustrates the opportunity cost of delay but is not a forecast of any drug's actual revenue or investment return.

3.4 Patents, prices, and access

Question 3.11. What is a pharmaceutical patent, and how does it differ from regulatory exclusivity?

A patent is a time-limited legal right over a qualifying invention's claims. It may enable the holder to prevent unauthorized commercial practice of those claims, subject to governing law. A pharmaceutical patent might cover an active compound, manufacturing process, formulation or qualifying use. **Regulatory exclusivity** is a separate rule that may restrict approval of certain competing applications for a specified time. Patent terms and regulatory exclusivities can overlap but are legally distinct. Patents are not themselves regulatory approval, and approval does not guarantee patent protection.

Question 3.12. Analyze how patents and induced innovation encourage pharmaceutical research while creating challenges for medicine affordability.

Patent incentives. Discovering a medicine involves large upfront costs, uncertain success and possible imitation after the discovery is disclosed. Temporary patent protection can let an inventor retain a larger share of potential sales revenue, improving the expected return on research and making some projects financially viable.
Affordability tension. Temporary market power may permit prices substantially above the marginal cost of manufacturing a medicine. Some patients, insurers or governments may face high expenses, and some beneficial treatments may not reach every patient who needs them. Generic or biosimilar competition after relevant protections and approval requirements permit entry can moderate prices.
Induced innovation. Firms may invest more in therapies when the number of potential patients, expected purchases or reimbursed demand rises. A larger market increases expected returns, holding research cost, margins and success probability fixed. Research therefore responds in part to economic demand, not solely to scientific need.
Policy trade-off. Society benefits from future drug discovery but also values current affordability. Time-limited intellectual property, public support for neglected research, procurement, competition policies and appropriate reimbursement can help balance research incentives with patient access. Profitability and social health value need not coincide.

Question 3.13. Why is the patent-filing date different from the market-entry date?

A pharmaceutical company may seek patent protection early in the research process, then spend years completing trials and regulatory review. Thus a portion of the patent term can pass before the medicine is marketed. In the United States, the ordinary term for many patents is generally measured from a relevant filing date, subject to legal qualifications. Actual enforceable protection depends on claims, filing history, adjustments, extensions, litigation and separate regulatory rules. These details vary across jurisdictions.

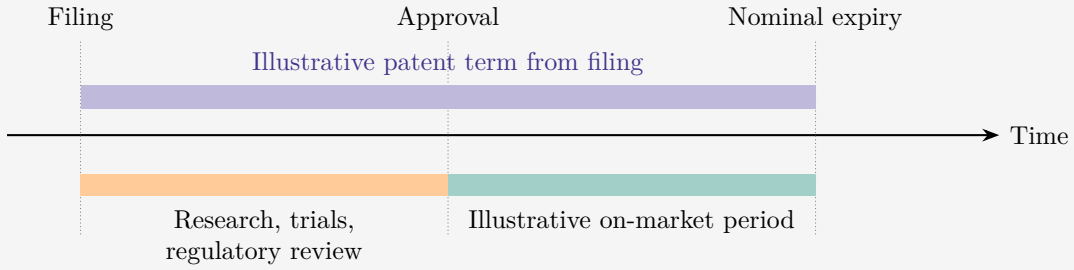


Figure 3.2: Filing, approval, and patent expiry are distinct dates.

Question 3.14. Derive the monopoly quantity and price for a patent-protected medicine under linear demand.

Assume one producer faces a downward-sloping inverse demand curve and constant marginal production cost. Total revenue equals the selling price times the number of units sold. Substituting inverse demand into revenue, then differentiating with respect to quantity, yields marginal revenue. A profit-maximizing producer sets marginal revenue equal to marginal cost. Substitute the resulting quantity back into demand to obtain the monopoly

price.

$$
\begin{aligned}
P(Q) &= a - bQ, \quad a > c > 0, \quad b > 0,\\
TC(Q) &= cQ + F, \qquad MC(Q) = c,\\
TR(Q) &= P(Q)Q = (a - bQ)Q\\
&= aQ - bQ^2,\\
MR(Q) &= \frac{dTR}{dQ} = a - 2bQ,\\
\pi(Q) &= TR(Q) - TC(Q) = aQ - bQ^2 - cQ - F,\\
\pi'(Q) &= a - 2bQ - c = 0,\\
2bQ_M &= a - c,\\
Q_M &= \frac{a-c}{2b}.\\
P_M &= a - bQ_M\\
&= a - b\left(\frac{a-c}{2b}\right)\\
&= a - \frac{a-c}{2}\\
&= \frac{2a - a + c}{2},\\
P_M &= \frac{a+c}{2}.
\end{aligned}
$$

The fixed research cost does not appear in the marginal condition because it is independent of output. It matters greatly for the decision to undertake research in the first place. This simple monopoly benchmark omits negotiated prices, insurance coverage, regulation, price discrimination and multi-firm competition. The second derivative of profit is negative, so the resulting stationary point is a maximum.

$$
\begin{aligned}
\pi''(Q) &= -2b < 0,\\
P_C &= c,\\
c &= a - bQ_C,\\
bQ_C &= a - c,\\
Q_C &= \frac{a-c}{b} = 2Q_M > Q_M.
\end{aligned}
$$

The model predicts higher prices and lower output under monopoly than in a price-equals-marginal-cost benchmark. It illustrates the static access cost of market power, while the possibility of recovering fixed R&D investment is a dynamic incentive for future research.

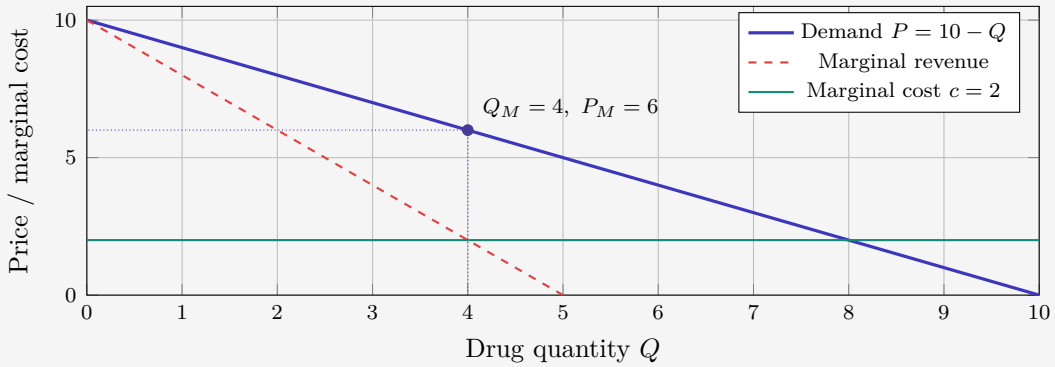


Figure 3.3: Demand, marginal revenue, and marginal cost under patent protection.

Question 3.15. Explain what changes when legally approved generics or biosimilars enter.

When applicable exclusivities cease to bar entry and competing products meet regulatory requirements, rivals can compete with the originator. For many small-molecule drugs, approved generics expand supply and can reduce transaction prices; biologic medicines instead generally face biosimilar competition under their applicable rules. Competition may lower the cost of care or enable additional patients to be treated. Prices are not guaranteed to equal marginal cost because market concentration, purchasing arrangements, distribution and regulation still matter.

3.5 Induced innovation and the market-size incentive

Question 3.16. Define induced pharmaceutical innovation and derive the private research threshold.

Induced innovation means that expected demand and financial incentives influence which scientific opportunities firms pursue. Suppose a potential therapy has a fixed development cost, a known contribution margin per successfully served patient, and a specified probability of commercial success. If it succeeds, a larger eligible patient market yields more expected contribution revenue. The firm chooses the project if expected net return is nonnegative.

$$
\begin{aligned}
F > 0 &= \text{fixed R\&D investment},\\
M &= \text{potential patient market},\\
m > 0 &= \text{contribution margin per served patient},\\
p \in (0,1] &= \text{probability of commercial success},\\
\mathbb{E}[\Pi] &= pMm - F,\\
\frac{\partial \mathbb{E}[\Pi]}{\partial M} &= pm > 0,\\
\mathbb{E}[\Pi] \geq 0 &\iff pMm - F \geq 0,\\
pMm &\geq F,\\
M &\geq M^* = \frac{F}{pm}.
\end{aligned}
$$

The expected-return line slopes upward in market size because each additional patient increases possible sales when other variables are held fixed. The threshold divides markets that do not cover the fixed investment in expectation from those that do. This is a deliberately simplified model; actual firms consider competitors, timing, trial evidence, capacity, regulation and multiple indications.

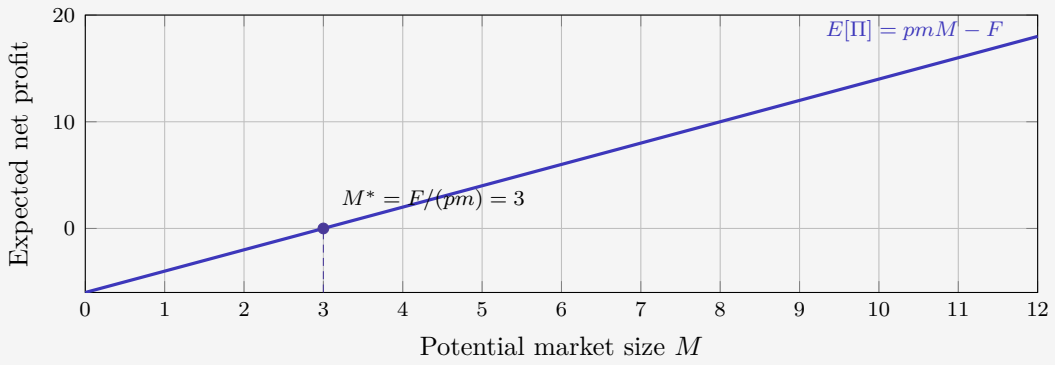


Figure 3.4: Expected return rises with market size ($pm = 2$, $F = 6$).

Question 3.17. Why might the disease with the highest social burden attract less private innovation?

Private firms respond to expected payments, costs, competition and success probabilities, not directly to the total health harm caused by a disease. A condition affecting many low-income patients may have high social value but limited purchasing power, while a smaller high-income market may support substantial expected revenue. Knowledge created by one research project can also help other researchers without generating revenue for the original firm. This gap between private and social benefits can justify public research grants, targeted incentives, advance purchasing, prizes and support for neglected diseases. A growing elderly population can alter the commercial market for therapies aimed at age-related conditions, although demand growth alone never guarantees scientific success.

$$\text{Private return} \neq \text{Social return} \qquad \text{in general.}$$

3.6 Drug regulation as an information problem

Question 3.18. Why do governments regulate the safety, effectiveness and quality of medicines?

Patients often cannot verify the ingredients, effectiveness, manufacturing quality or delayed side effects of medical products. Regulators therefore require evidence and monitor risks. Approval standards can protect patients from ineffective or harmful medicines and support public confidence. They can also impose substantial trials, documentation and compliance costs. Delayed access to a genuinely beneficial medicine may carry an opportunity cost, so the objective is credible, proportionate protection rather than either no regulation or unlimited delay. Product approval, drug pricing and insurance reimbursement are different policy decisions.

Question 3.19. Why does regulation continue after a medicine has been approved?

A clinical trial cannot expose every possible adverse effect because it studies a limited population for a limited time. Rare events or interactions may become visible only after many patients use the drug. Postmarketing reports and further studies can justify new

warnings, additional evidence requirements, narrower indications or withdrawal in serious cases. Continued learning also provides evidence on additional indications, groups and long-term effects. Approval is an evidence-based decision at one point in time, not proof that all uncertainty has disappeared.

3.7 Medical technology, effectiveness, and prices

Question 3.20. What counts as medical technology, and why are sticker price and value different?

Medical technology includes pharmaceuticals, vaccines, imaging, diagnostics, monitoring equipment, surgical techniques, prostheses and digital methods of delivering care. A technology can cost more per episode while preventing later expensive complications or generating a greater improvement in health. Consequently, analysts distinguish the payment for a service, the resource cost per successfully treated person, and the cost per unit of health improvement. The cost per unit of health benefit may fall even when the cost per treatment rises.

Question 3.21. Calculate average cost per health gain and the incremental cost-effectiveness ratio for two illustrative therapies.

The lecture compares an older therapy and a newer therapy. The newer therapy costs more in total but produces a larger gain in quality-adjusted life years. First divide each treatment's total resource cost by its health gain to obtain average cost per gain. When one treatment replaces another, the more relevant calculation compares *additional* cost with *additional* health gain, rather than comparing average ratios alone.

Treatment	Resource cost (US$)	Health gain (QALYs)
Older	100	0.5
Newer	150	1.5

$$\begin{aligned}
\text{old average cost per QALY} &= \frac{100}{0.5} = 200 \text{ US\$/QALY},\\
\text{new average cost per QALY} &= \frac{150}{1.5} = 100 \text{ US\$/QALY},\\
\Delta C &= C_{\text{new}} - C_{\text{old}} = 150 - 100 = 50,\\
\Delta H &= H_{\text{new}} - H_{\text{old}} = 1.5 - 0.5 = 1.0,\\
ICER &= \frac{\Delta C}{\Delta H} = \frac{C_{\text{new}} - C_{\text{old}}}{H_{\text{new}} - H_{\text{old}}},\\
ICER &= 50 \text{ US\$ per additional QALY}.
\end{aligned}$$

The newer treatment costs fifty additional dollars and produces one additional quality-adjusted life year in this artificial comparison. Whether it should be adopted depends on opportunity costs, the available budget, the clinical population and the decision-making framework. These computations do not represent observed therapies.

3.8 Technology and aggregate medical expenditure

Question 3.22. Derive how health expenditure depends on price and quantity and obtain its growth decomposition.

For a single well-defined medical service, expenditure equals price per treatment times number of treatments. Total expenditure may increase if utilization grows sufficiently quickly even when unit price falls. To study growth over time, differentiate the expenditure identity using the product rule, then divide by total expenditure. The result expresses proportional expenditure growth as the sum of proportional price growth and proportional quantity growth.

$$\begin{aligned} E(t) &= P(t)Q(t), \\ \frac{dE}{dt} &= \frac{d}{dt}[P(t)Q(t)], \\ &= Q(t)\frac{dP}{dt} + P(t)\frac{dQ}{dt}, \\ \frac{1}{E}\frac{dE}{dt} &= \frac{Q}{PQ}\frac{dP}{dt} + \frac{P}{PQ}\frac{dQ}{dt}, \\ &= \frac{1}{P}\frac{dP}{dt} + \frac{1}{Q}\frac{dQ}{dt}, \\ \frac{d\ln E}{dt} &= \frac{d\ln P}{dt} + \frac{d\ln Q}{dt}. \end{aligned}$$

The final equality is exact for instantaneous growth of a single well-defined service with positive price and quantity. When there are multiple treatments, total spending is the sum of price times quantity across those treatments. Shifts toward more intensive or different types of care also matter, so one aggregate price and one aggregate quantity can hide important changes in the service mix.

$$E_{\text{total}} = \sum_{j=1}^{J} P_j Q_j.$$

Question 3.23. Show numerically how total medical spending can rise even when treatment price falls.

The lecture considers a treatment whose unit price falls after an innovation while the number of treated patients rises. Multiply the original price by the original number treated and the new price by the new number treated. Next calculate the percentage changes. The result demonstrates that expansion of utilization can outweigh the savings per treatment.

Measure	Before innovation	After innovation
Price per treatment (US$)	100	80
Number of treatments	1,000	1,500
Total expenditure (US$)	100,000	120,000

$$
\begin{aligned}
E_0 &= P_0 Q_0 = 100(1{,}000) = 100{,}000, \\
E_1 &= P_1 Q_1 = 80(1{,}500) = 120{,}000, \\
\%\Delta P &= \frac{80 - 100}{100} \times 100 = -20\%, \\
\%\Delta Q &= \frac{1{,}500 - 1{,}000}{1{,}000} \times 100 = 50\%, \\
\%\Delta E &= \frac{120{,}000 - 100{,}000}{100{,}000} \times 100 = 20\%.
\end{aligned}
$$

The unit price falls by twenty percent, the volume of treatment rises by fifty percent, and total spending grows by twenty percent. These percentage changes need not add exactly in a discrete two-period calculation because the price–quantity interaction matters. Innovation can therefore improve affordability per treatment yet still increase the aggregate health-care budget.

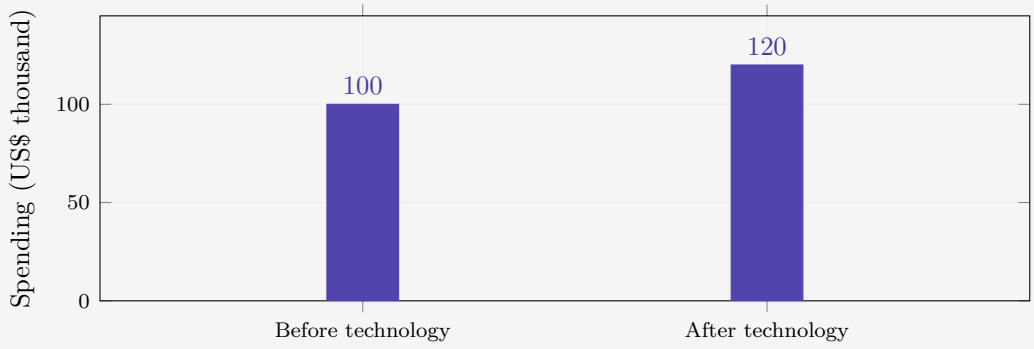


Figure 3.5: Total expenditure rises from US$100,000 to US$120,000 despite a lower unit price.

Question 3.24. Explain why a new technology may increase demand even if its direct production cost falls.

A safer, more convenient or more effective therapy may become attractive to people who previously did not seek treatment. Clinicians may recommend it for additional conditions, more frequent monitoring or new patient groups. Coverage can further lower the amount a patient pays at the point of treatment and make more patients willing or able to use the service. Thus an innovation may shift the demand curve outward as well as change its supply cost. The net effect on expenditure depends on the change in both price and quantity.

$$
\begin{aligned}
&P_{\text{patient}} = \alpha P, \qquad 0 \le \alpha \le 1, \\
&\quad \alpha \downarrow \implies P_{\text{patient}} \downarrow \implies Q_D \uparrow \quad \text{if demand slopes downward.}
\end{aligned}
$$

3.9 Cost effectiveness, budget impact, and policy design

Question 3.25. Why can a cost-effective medicine nevertheless be difficult to afford?

Cost effectiveness compares added health benefit with the added resources required for one patient or a comparable unit of treatment. Budget impact asks how much total financing will be needed when the treatment is offered to everyone who may use it. A medicine can have attractive incremental value yet place a large aggregate burden on a public budget or insurance fund if many people become eligible. Both considerations matter for sustainable coverage decisions.

$$\begin{aligned} \Delta B &\approx N_{\text{eligible}} u (C_{\text{new}} - C_{\text{old}}), \\ N_{\text{eligible}} &= 1{,}000{,}000, \qquad u = 0.5, \\ C_{\text{new}} - C_{\text{old}} &= 200 \text{ US\$}, \\ \Delta B &\approx 1{,}000{,}000(0.5)(200) \\ &= 100{,}000{,}000 \text{ US\$}. \end{aligned}$$

The calculation assumes a fixed eligible population, uptake rate and incremental cost per treated person while ignoring other changes and offsets. It shows why a treatment may justify its extra cost per patient while requiring substantial new total funding.

Question 3.26. Identify pathways through which medical innovation can increase or decrease health-care expenditure.

Cost-reducing channels include cheaper tests with similar accuracy, less invasive procedures, effective prevention of costly complications, and lower production costs. **Expenditure-increasing channels** include treating previously untreatable conditions, expanding eligibility, more intensive diagnostics, expensive new products, repeated procedures and longer survival accompanied by additional future care. Insurance coverage, provider payment and purchasing arrangements influence how fast a technology spreads. Its net effect on total expenditure is therefore an empirical question rather than a theoretical certainty.

Question 3.27. Which policy tools can support innovation, safety, clinical value and affordable access?

Possible tools include publicly funded research in socially important areas, time-limited patents followed by legally permitted competition, strong evidence requirements and postmarket monitoring, generic and biosimilar pathways, procurement and negotiated purchasing, health-technology assessment, and reimbursement designed to reward clinical benefit. Policy must recognize several distinct problems: insufficient research incentives, information gaps about safety and effectiveness, market power in the price of an existing medicine, and the fiscal consequences of broad adoption. A single instrument rarely resolves all four problems.